

ROS2 개발 환경 구축



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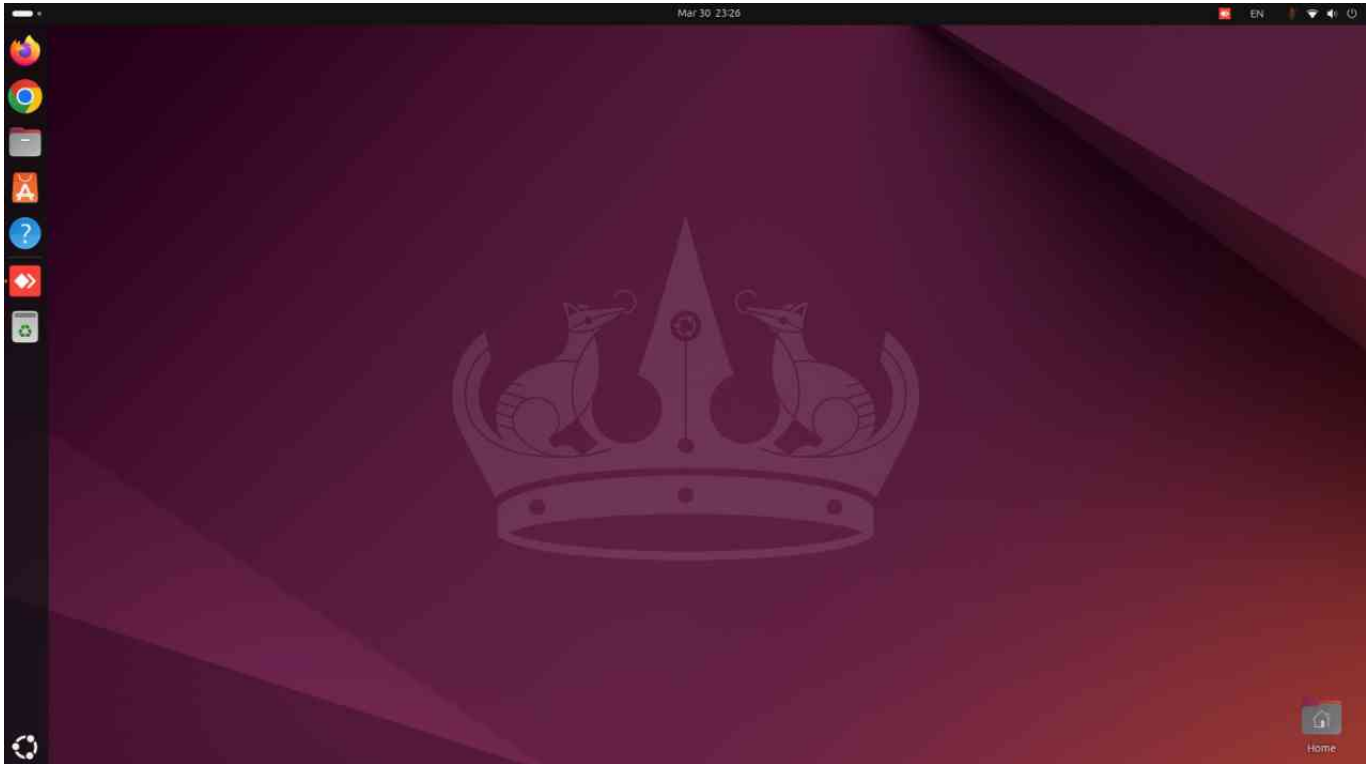
24 Colcon 패키지 설치

1 영상으로.(영상은 Humble 버전임에 유의!)

ROS2 개발 환경 구축

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우리는 우분투 24.04에서 시작합니다.



ROS2 Jazzy 설치 페이지

<https://docs.ros.org/en/jazzy/Installation.html> (<https://docs.ros.org/en/jazzy/Installation.html>)

Debian Packages (Ubuntu) 선택

Installation — ROS 2 Document

docs.ros.org/en/jazzy/Installation.html

ROS 2 Documentation: Jazzy



Search docs

Installation

Ubuntu (deb packages)

Windows (binary)

RHEL (RPM packages)

Alternatives

Maintain source checkout

Testing with pre-release binaries

DDS implementations

Other Versions

v: jazzy

/ Installation

Edit on GitHub

Installation

Options for installing ROS 2 Jazzy Jalisco:

Binary packages

Binaries are only created for the Tier 1 operating systems listed in [REP-2000](#). If you are not running any of the following operating systems you may need to build from source or use a [container solution](#) to run ROS 2 on your platform.

We provide ROS 2 binary packages for the following platforms:

- Ubuntu Linux - Noble Numbat (24.04)
 - **deb packages (recommended)**
 - [binary archive](#)
- RHEL 9
 - [RPM packages](#) (recommended)
 - [binary archive](#)

설치 과정 따라가기

Ubuntu (deb packages) — RO

docs.ros.org/en/jazzy/installation/Ubuntu-Install-Debs.html

ROS 2 Documentation: Jazzy



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 - Enable required repositories
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- Install ROS 2
 - Install additional RMW implementations (optional)
- Setup environment
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- Troubleshoot
- Uninstall

Deb packages for ROS 2 Jazzy Jalisco are currently available for Ubuntu Noble (24.04). The target platforms are defined in [REP 2000](#).

Resources

- Status Page:
 - ROS 2 Jazzy (Ubuntu Noble 24.04): [amd64](#), [arm64](#)
- [Jenkins Instance](#)
- [Repositories](#)

System setup

Set locale

Make sure you have a locale which supports [UTF-8](#). If you are in a minimal environment (such as a docker container), the locale may be something minimal like [POSIX](#). We test with the following settings. However, it should be fine if you're using a different UTF-8 supported locale.

Set locale

System setup

Set locale

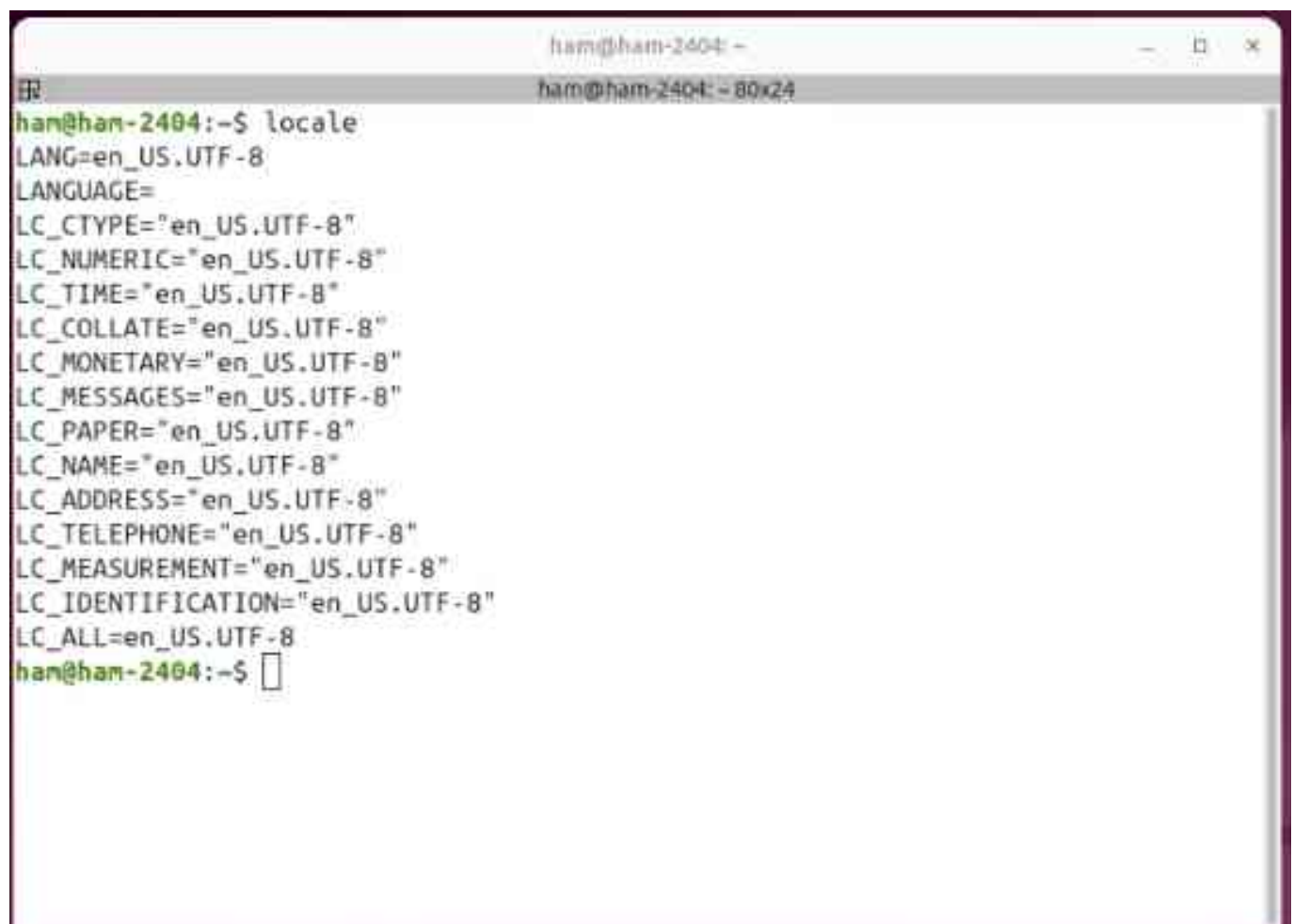
Make sure you have a locale which supports `UTF-8`. If you are in a minimal environment (such as a docker container), the locale may be something minimal like `POSIX`. We test with the following settings. However, it should be fine if you're using a different UTF-8 supported locale.

```
locale: # check for UTF-8

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale: # verify settings
```

locale 설정 확인

A terminal window titled 'ham@ham-2404: ~' with a subtitle 'ham@ham-2404: ~ 80x24'. The user has entered the command 'locale' and the terminal displays the following output:

```
ham@ham-2404:~$ locale
LANG=en_US.UTF-8
LANGUAGE=
LC_CTYPE="en_US.UTF-8"
LC_NUMERIC="en_US.UTF-8"
LC_TIME="en_US.UTF-8"
LC_COLLATE="en_US.UTF-8"
LC_MONETARY="en_US.UTF-8"
LC_MESSAGES="en_US.UTF-8"
LC_PAPER="en_US.UTF-8"
LC_NAME="en_US.UTF-8"
LC_ADDRESS="en_US.UTF-8"
LC_TELEPHONE="en_US.UTF-8"
LC_MEASUREMENT="en_US.UTF-8"
LC_IDENTIFICATION="en_US.UTF-8"
LC_ALL=en_US.UTF-8
ham@ham-2404:~$
```

copy해서 터미널에 붙여넣기 (sudo 조심)

```
locale # check for UTF-8

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale # verify settings
```

Ubuntu 자체가 제공하는 “universe” 저장소 활성화

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

ROS 2 전용 APT 저장소를 자동으로 등록

```
$ sudo apt update && sudo apt install curl -y
$ export ROS_APT_SOURCE_VERSION=$(curl -s https://api.github.com/repos/ros-infrastructure/ros-apt-source/releases/latest | grep -F "tag_name" | cut -d '"' -f 2)
$ curl -L -o /tmp/ros2-apt-source.deb "https://github.com/ros-infrastructure/ros-apt-source/releases/download/${ROS_APT_SOURCE_VERSION}/ros2-apt-source.deb"
$ sudo apt install /tmp/ros2-apt-source.deb
```

apt update 진행


```
ham@ham-2404: ~  
ham@ham-2404: ~ 91x45  
ham@ham-2404:~$ sudo apt update  
[sudo] password for ham:  
Get:1 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]  
Get:2 https://packages.microsoft.com/repos/code stable/main armhf Packages [18.1 kB]  
Get:3 https://packages.microsoft.com/repos/code stable/main amd64 Packages [17.8 kB]  
Get:4 https://packages.microsoft.com/repos/code stable/main arm64 Packages [18.0 kB]  
Get:5 https://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]  
Get:6 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:7 http://deb.anydesk.com all InRelease [7,386 B]  
Get:8 https://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,213 B]  
Hit:9 http://archive.ubuntu.com/ubuntu noble InRelease  
Err:7 http://deb.anydesk.com all InRelease  
The following signatures couldn't be verified because the public key is not available: NO  
_PUBKEY A2FB21D5A8772835  
Get:11 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:12 http://packages.ros.org/ros2/ubuntu noble InRelease [4,676 B]  
Get:13 http://packages.ros.org/ros2/ubuntu noble/main amd64 Packages [1,076 kB]  
Get:14 https://packagecloud.io/slacktechnologies/slack/debian jessie InRelease [29.1 kB]  
Get:15 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [711 kB]  
Get:16 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:17 https://packagecloud.io/slacktechnologies/slack/debian jessie/main amd64 Packages [4,027 B]  
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [960 kB]  
Get:19 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [136 kB]  
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [8,996 B]  
Get:21 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [7,068 B]  
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [810 kB]  
Get:23 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [164 kB]  
Get:24 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]  
Get:25 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [468 B]  
Get:26 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [822 kB]  
Get:27 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [177 kB]  
Get:28 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [51.9 kB]  
Get:29 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]  
Get:30 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.6 kB]  
Get:31 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3,792 B]  
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [213 kB]  
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [151 kB]  
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [13.5 kB]  
Get:35 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [842 kB]  
Get:36 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]  
Get:37 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
```

apt upgrade 진행

Code

```
1 sudo apt upgrade
```

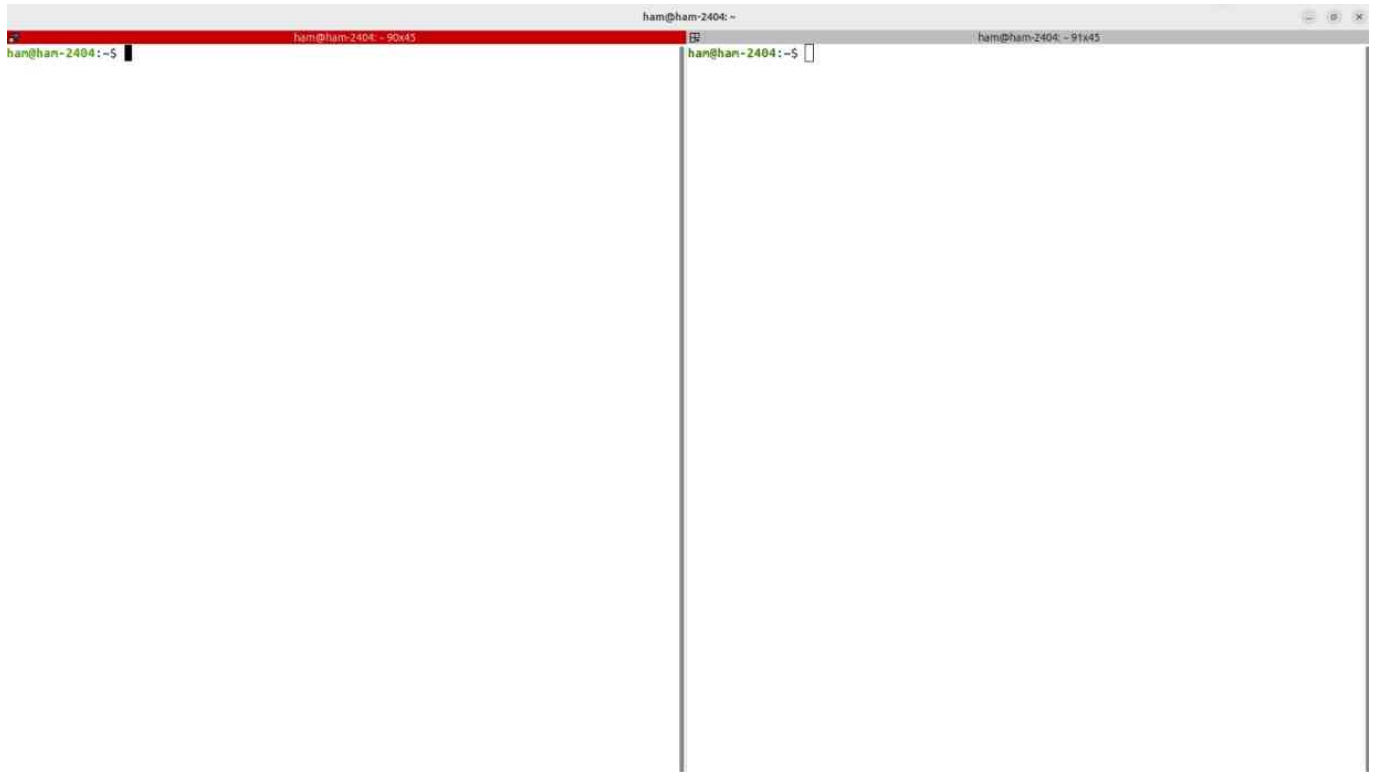
ros-jazzy-desktop 설치 시작

```
sudo apt install ros-jazzy-desktop
```

설치 중

```
sip-dev tango-icon-theme tcl-dev tcl8.6-dev tk-dev tk8.6-dev  
uncrustify unicode-data unixodbc-common unixodbc-dev uuid-dev  
va-driver-all vdpau-driver-all vtk9 x11proto-dev  
xorg-sgml-doctools xtrans-dev  
0 upgraded, 895 newly installed, 0 to remove and 4 not upgraded.  
Need to get 516 MB of archives.  
After this operation, 2,383 MB of additional disk space will be used.  
Do you want to continue? [Y/n] y
```

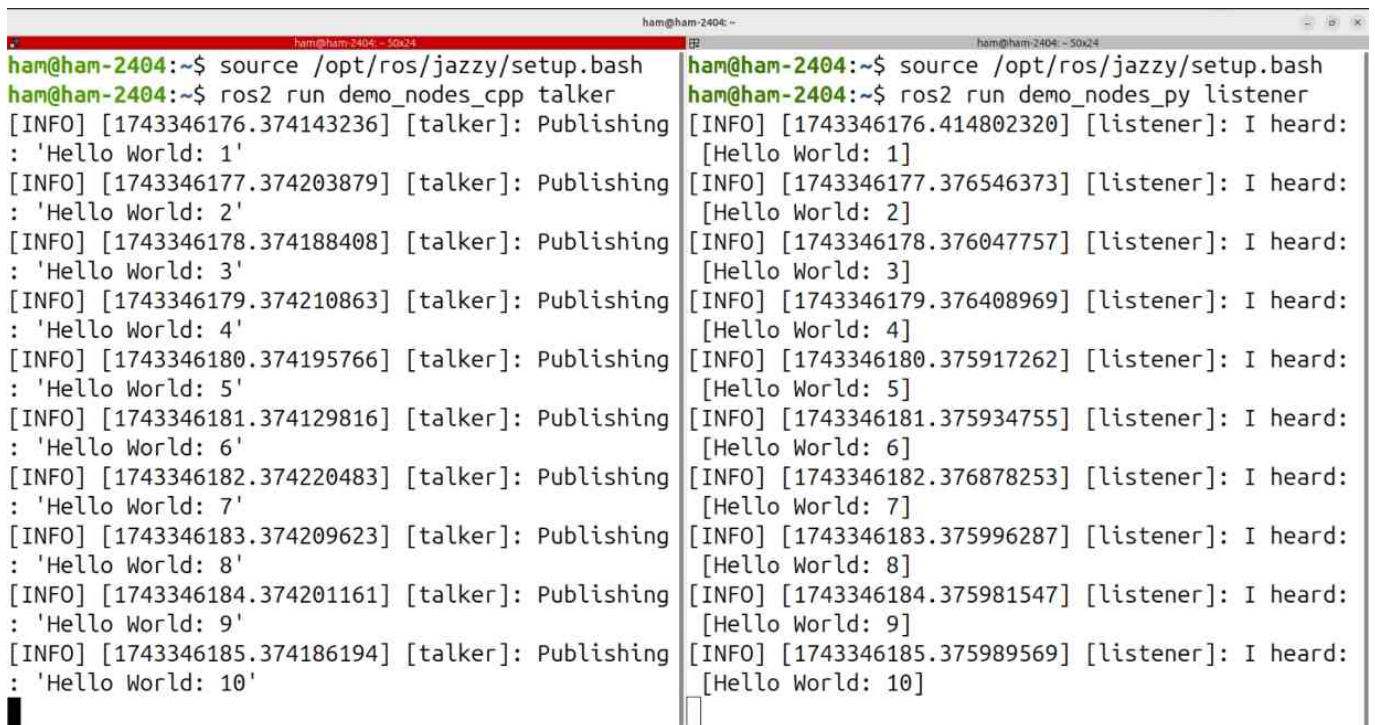
설치 후 두 개의 터미널 준비해서



둘다 ROS2 Jazzy의 setup.bash를 source로 부르고

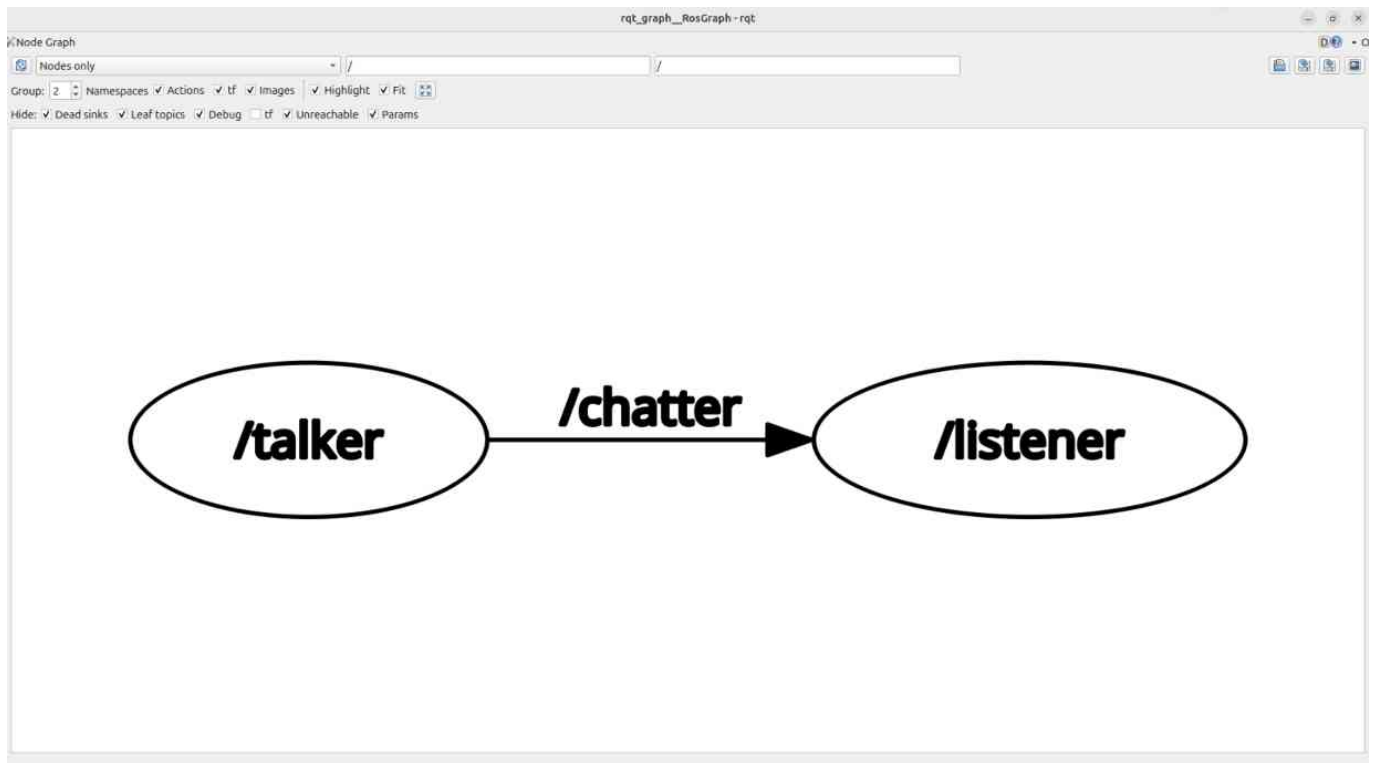


각각 talker와 listener 실행



잘 작동되면 성공적으로 설치된 것 입니다.

이렇게 연결된 것 입니다.

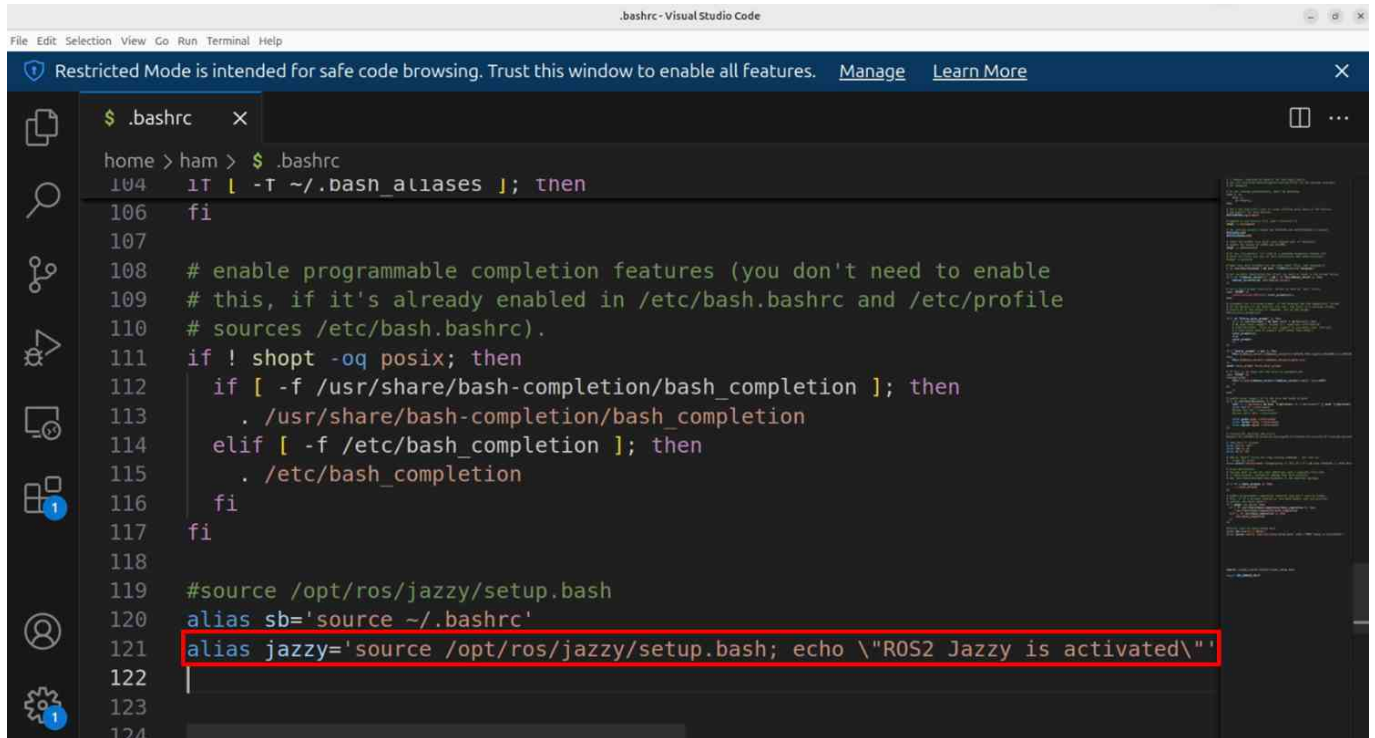


터미널이 실행될 때마다 ROS2 환경을 불러오려면?

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

~/.bashrc 파일의 내용은 터미널이 실행될 때 자동으로 실행

혹은 alias로 필요할 때마다 불러오게 설정



```
.bashrc
home > ham > $ .bashrc
104 1T | -T ~/.bash_aliases ]; then
106 fi
107
108 # enable programmable completion features (you don't need to enable
109 # this, if it's already enabled in /etc/bash.bashrc and /etc/profile
110 # sources /etc/bash.bashrc).
111 if ! shopt -oq posix; then
112     if [ -f /usr/share/bash-completion/bash_completion ]; then
113         . /usr/share/bash-completion/bash_completion
114     elif [ -f /etc/bash_completion ]; then
115         . /etc/bash_completion
116     fi
117 fi
118
119 #source /opt/ros/jazzy/setup.bash
120 alias sb='source ~/.bashrc'
121 alias jazzy='source /opt/ros/jazzy/setup.bash; echo \"ROS2 Jazzy is activated\";'
122
123
124
```

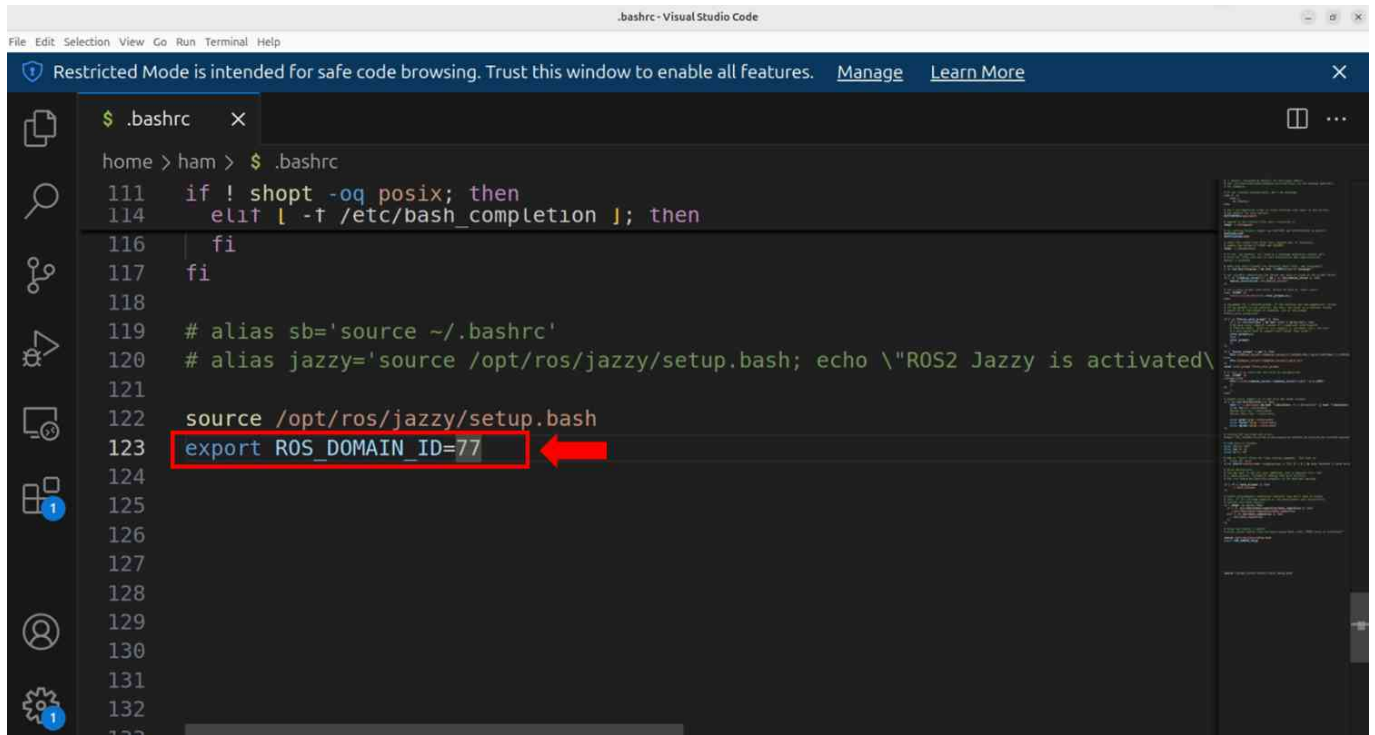
터미널을 새로 열 때마다 등록해둔 단축어를 사용



```
ham@ham-2404: ~
ham@ham-2404: ~ 91x43
ham@ham-2404:~$ jazzy
"ROS2 Jazzy is activated"
ham@ham-2404:~$
```

ROS2_DOMAIN_ID

- 서로의 데이터를 공유하면 문제가 생길 수 있으므로 격리시켜야 한다!



```
.bashrc
home > ham > $ .bashrc
111 if ! shopt -oq posix; then
114     elif [ -f /etc/bash_completion ]; then
116         fi
117     fi
118
119 # alias sb='source ~/.bashrc'
120 # alias jazzy='source /opt/ros/jazzy/setup.bash; echo \"ROS2 Jazzy is activated\"'
121
122 source /opt/ros/jazzy/setup.bash
123 export ROS_DOMAIN_ID=77
124
125
126
127
128
129
130
131
132
133
```

추가 패키지 설치

rosdep 등 개발에 필요한 패키지를 설치

```
sudo apt update && sudo apt install ros-dev-tools
```

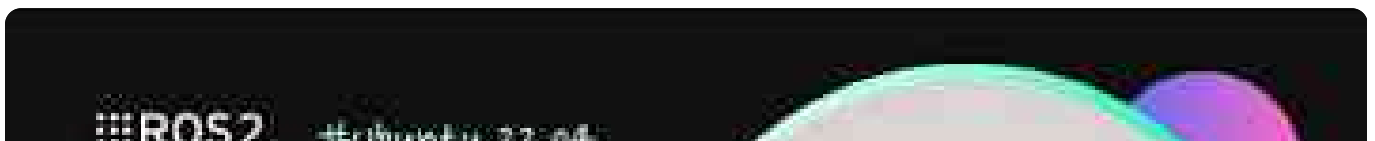
Colcon 패키지 설치

직접 패키지를 개발하고 빌드하기 위해 필요

```
sudo apt install python3-colcon-common-extensions
```

영상으로.

(영상은 Humble 버전임에 유의!)



Rush To ROS

#Humble 화동영님

Ubuntu 설치 및 설정



무작정 따라하는 ROS2 - 입문과정

https://youtu.be/KR_A_gjzSmY



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